# The translator’s signal, measured

## Degree Zero of Translation: Using Interlinear Baselines to Quantify Translator Intervention

**Maciej Rapacz** · **Aleksander Smywiński-Pohl**

AGH University of Kraków

mrapacz.com/eacl2026

### BACKGROUND

Literary translation is an act of rewriting: every translator restructures, shifts meaning, and adapts style. Translation studies classifies these choices qualitatively — literal, formal, dynamic, paraphrase — but existing approaches lack a scalable method to measure how far a translation departs from its source.

### METHOD

An interlinear translation — a word-for-word gloss preserving source syntax — serves as our baseline of zero intervention. We embed each chapter of a translation and its interlinear counterpart using Qwen3-Embedding-8B, then compute the Intervention Vector:

$$\boldsymbol{V}_{\text{intervention}} = \boldsymbol{V}_{\text{translation}} - \boldsymbol{V}_{\text{interlinear}}$$

Its magnitude (L2 norm) measures the extent of departure: small for literal renderings, large for paraphrases.

### DATASET

74 New Testament translations from the targum corpus, across 5 languages, each aligned chapter-by-chapter (260 chapters) with a language-specific interlinear baseline.

| Language | Literal | Formal | Dynamic | Paraphrastic | Total |
|---|---|---|---|---|---|
| English | 3 | 4 | 7 | 2 | 16 |
| French | 1 | 6 | 4 | 3 | 14 |
| Italian | 4 | 4 | 2 | 2 | 12 |
| Polish | 4 | 7 | 2 | 3 | 16 |
| Spanish | 3 | 2 | 5 | 6 | 16 |
| **Total** | **15** | **23** | **20** | **16** | **74** |

### KEY CONTRIBUTION

An unsupervised, reference-free method that recovers the translation strategy spectrum using interlinear translation as baseline.

### KEY FINDINGS

#### Spectrum recovery

Across all 5 languages, literal and formal strategies yield significantly shorter vectors than dynamic and paraphrase. In English and French, all four groups separate fully ($p < 0.001$).

#### Pairwise sensitivity

At the individual translation level, 87% of English pairs show significant differences in intervention magnitude (63% French, 41% Polish, 36% Italian, 35% Spanish).

#### Topology

PCA projections reveal that translations with similar magnitude can depart in different directions. In English, the First Nations Version and the Orthodox Jewish Bible both show high intervention but along orthogonal axes — one domesticating, the other foreignizing:

Acts 17:33

Interlinear: Thus Paul went out from the midst of them
FNV: So Small Man went on his way.
OJB: Thus did Rav Sha’ul go out from the midst of them.

### THE TRANSLATION SPECTRUM

The same verse at four levels of intervention:

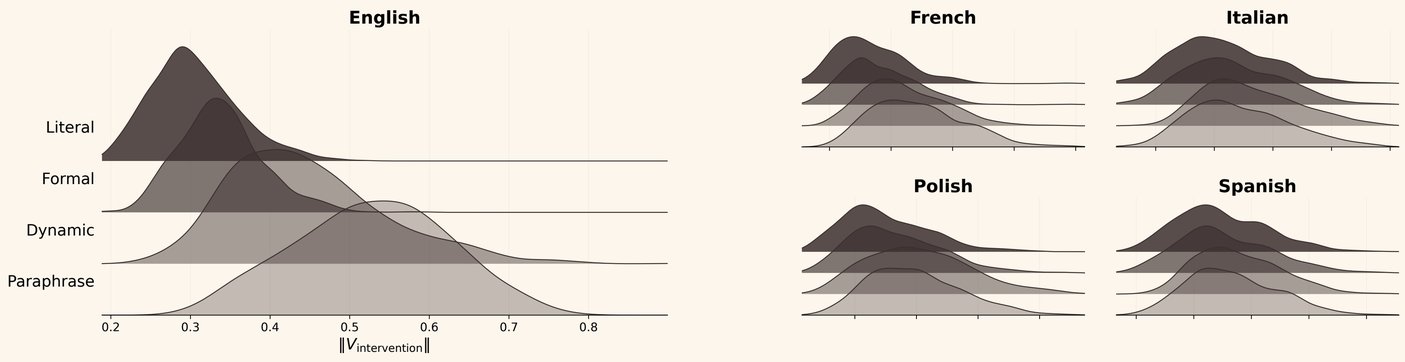
Greek source · SBLGNT · Matthew 13:9

| ὁ | ἔχων | ὦτα | ἀκουέτω |
|---|---|---|---|
| *The [one]* | *having* | *ears* | *let him hear* |

| | | Dist. |
|---|---|---|
| **LITERAL** | | |
| DLNT | Let the one having ears, hear". | 0.547 |
| **FORMAL** | | |
| NRSVue | Let anyone with ears listen!" | 0.647 |
| **DYNAMIC** | | |
| NLT | Anyone with ears to hear should listen and understand." | 0.656 |
| **PARAPHRASTIC** | | |
| MSG | "Are you listening to this? Really listening?" | 0.903 |

### DISTRIBUTION OF INTERVENTION MAGNITUDES

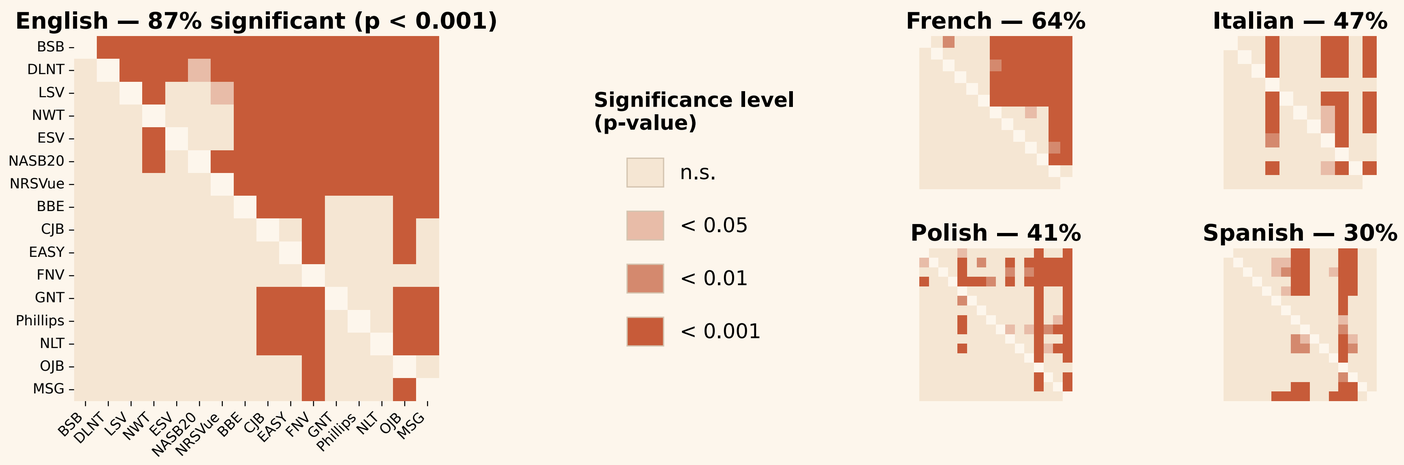
The intervention magnitude increases along the expected spectrum from literal to paraphrase:




Distribution of $\|V_{\text{intervention}}\|$ per strategy group. Literal strategies cluster low; paraphrases extend further.

### PAIRWISE SIGNIFICANCE

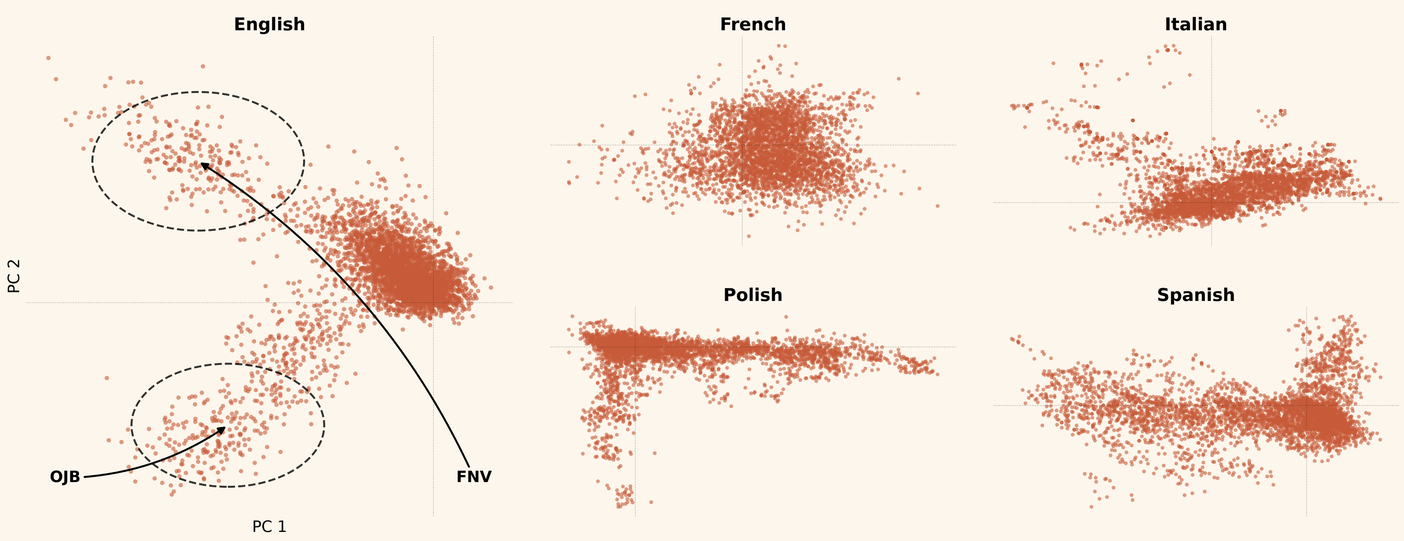
At the level of individual translations, most pairs show significant differences in intervention magnitude:




Mann–Whitney U (one-sided, Bonferroni-corrected). Color intensity indicates significance level.

### TOPOLOGY OF INTERVENTION

Projecting the intervention vectors onto a plane reveals that direction, not just magnitude, varies across translations:




PCA of intervention vectors. Baseline at origin. FNV and OJB depart along orthogonal axes.

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